

Master's Comprehensive Examination

Part II

Applied (90 minutes)

March 11, 2000

Instructions: Answer all four questions. Open book and open notes.

1. The data in the accompanying table were obtained through a survey of randomly selected subjects.

(a) If one of the survey subjects is randomly selected, find the probability of getting someone ticketed for speeding.

(b) If one of the survey subjects is randomly selected, find the probability of getting a male or someone ticketed for speeding.

(c) Find the probability of getting someone ticketed for speeding, given that the selected person is a man.

(d) Find the probability of getting someone ticketed for speeding, given that the selected person is a woman.

(e) Use a 0.05 level of significance to test the claim that the percentage of women ticketed for speeding is less than the percentage for men. Can we conclude that men generally speed more than women?

	Ticketed for Speeding Within the Last Year?	
	Yes	No
Men	26	224
Women	27	473

Based on data from R.H. Bruskin Associates.

2. The Allan Preparation Course is designed to help students achieve better scores on the SAT exam. The accompanying table lists results for randomly selected students. Use the indicated test with a 0.05 level of significance to test the hypothesis that the course has no effect on SAT scores.

- (a) Sign test.
- (b) Wilcoxon signed-rank test.
- (c) t test.

(d) How do the preceding results support the statement that nonparametric tests lack the sensitivity of parametric tests (so stronger evidence is required before a null hypothesis is rejected)?

Subject	A	B	C	D	E	F	G	H	I	J
SAT score before course	700	840	830	860	840	690	830	1180	930	1070
SAT score after course	800	840	820	980	980	800	800	1270	1080	1220

3. A marketing researcher studied annual sales of a product that had been introduced ten years ago. The data are as follows, where X is the year (coded from 0 - 9) and Y is sales in thousands of units:

i :	1	2	3	4	5	6	7	8	9	10
X_i :	0	1	2	3	4	5	6	7	8	9
Y_i :	98	135	162	178	221	232	283	300	374	395

- (a) Prepare a scatter plot of the data. Does a linear relation appear adequate here?
- (b) Use the transformation $Y' = \sqrt{Y}$ and obtain the estimated linear regression function for the transformed data.
- (c) Plot the estimated regression line and the transformed data. Does the regression line appear to be a good fit to the transformed data?
- (d) Express the estimated regression function in the original units.
- (e) Use the model obtained in (b) to predict the sales Y in the next year (the eleventh year after the product is introduced to the market). Obtain the prediction interval.

4. An owner of a chicken farm would like to test a new feed supplement that is supposed to increase egg production. He randomly selects 8 groups of 100 chickens each and feeds them various levels of the feed supplement. Listed are the amounts of feed supplement and the number of eggs per day.

Group	1	2	3	4	5	6	7	8
Feed supplement	10	10	15	15	20	20	25	25
Number of eggs	78	81	85	79	96	89	93	87

- Construct a scatterplot of the data. Does there appear to be an upward trend that would predict the number of eggs?
- Obtain the least squares estimates of β_0 and β_1 for the model $y = \beta_0 + \beta_1 x + \epsilon$.
- Test the hypothesis $\beta_1 = 0$. Do the results of this test indicate that a linear trend is significant?
- How many additional eggs are expected to be produced for each additional unit of feed supplement?
- Predict the number of eggs if the feed supplement is set at 20 units. Provide the corresponding confidence interval as well.
- Would this model be useful in predicting the number of eggs if the feed supplement is increased to 40 units? Explain.